AMRITA VISHWA VIDYAPEETHAM
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(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

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Problem Statement

Farmers frequently make crop selling decisions without reliable information about future demand, market prices, transportation costs, or local demand.

Problem Understanding

- Smallholder and mid-sized farmers, especially those far from major markets or without access to market data, cooperatives, or buyers.
- Weather is increasingly unpredictable and forecasts are too generic. Price and demand information is scattered, outdated, or controlled by middlemen. Few tools connect weather, soil, and market demand to recommend the right crop.
- Farmers face crop failure, oversupply of the wrong crops, low prices, and lost income. Buyers and local markets deal with supply gluts or shortages, and communities see higher food prices and more waste.

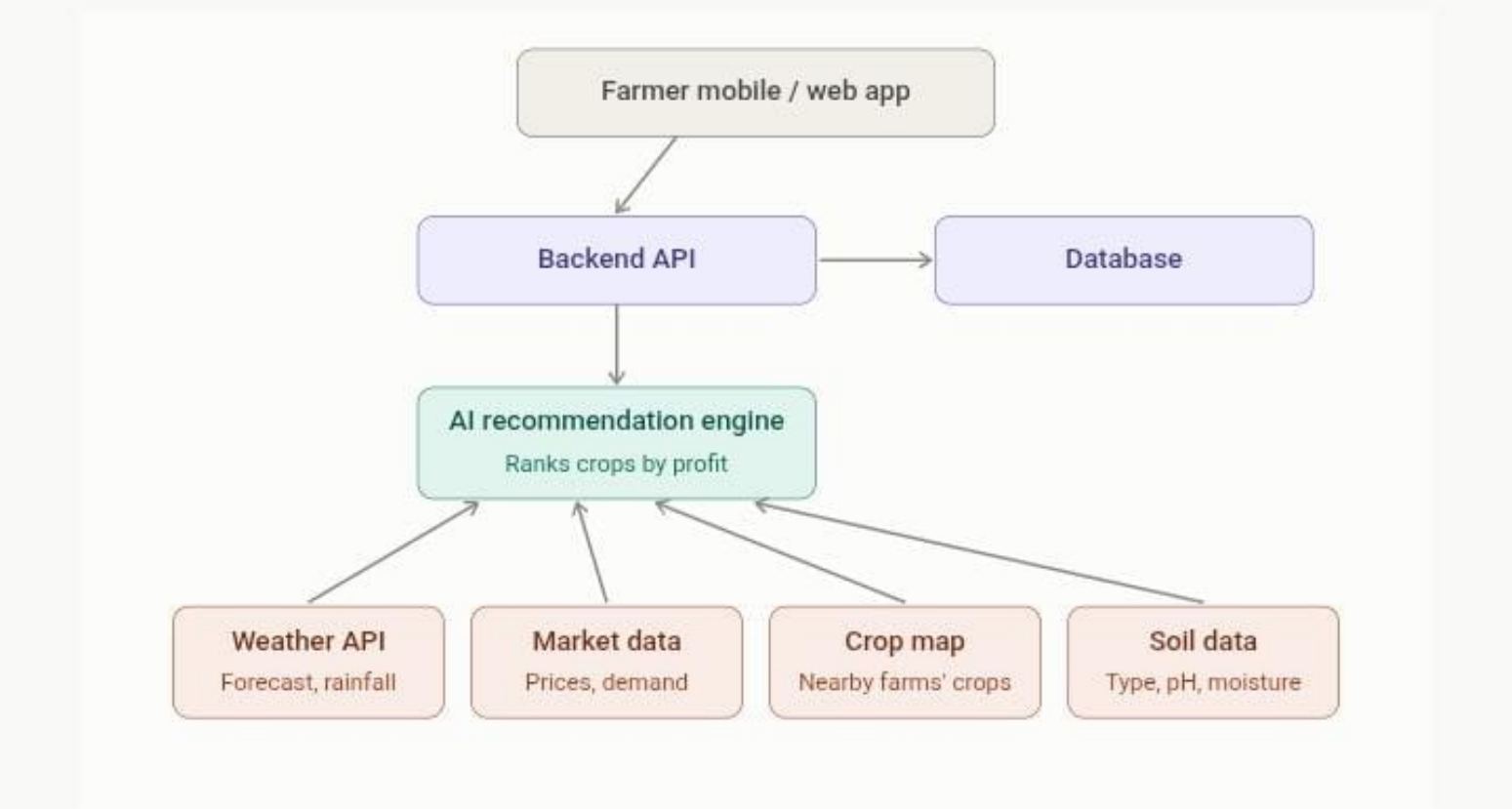
Existing Gaps & Challenges

- Fragmented tools. Existing apps cover either weather forecasts or market prices, but rarely combine both into one crop recommendation.
- Climate unpredictability. Due to global warming, weather patterns are shifting, so traditional farming knowledge and seasonal calendars no longer work, and farmers can't reliably predict rainfall or temperature.
- No accessible, affordable system links changing weather, soil, and real-time market demand to tell farmers what to plant and when to sell for the best profit.

Our Proposed Solution

- An AI-powered crop advisory app that analyzes weather, soil, market demand, and what surrounding farmers are growing, then recommends the crop with the highest profit and demand.
- The app takes the farmer's location and soil details, combines them with weather and climate forecasts, live market prices, and a geographical report of nearby crops. It then ranks crop options by expected profit and demand, favoring crops that are in demand but not oversupplied locally.
- Farmers get a clear, location-specific recommendation for a high-profit, in-demand crop, which reduces crop failure and oversupply and increases income, even as global warming makes weather harder to predict.

Technical Approach



The farmer enters location and soil details in the app. The backend passes this to the AI engine, which combines weather forecasts, market prices and demand, soil data, and nearby farms' crops. It then ranks crops by profit and demand and shows the best crop, planting time, and where to sell.

Tech Stack

Innovation & Uniqueness

- Existing apps show either weather or prices. Ours combines weather, soil, market demand, and a geographical report of what nearby farmers are growing into one crop recommendation
- Farmers struggle with unpredictable weather and unclear demand. Using AI on live, location-specific data gives them a clear answer on what to plant and when to sell, even as global warming makes weather harder to predict.
- Awareness of surrounding farms' crops. By recommending a high-profit, in-demand crop that is not oversupplied locally, we help farmers avoid crowded markets and earn more.

Feasibility & Scalability

- **Feasibility:** Built with low-cost resources: open-source tools (Python, React Native, PostgreSQL), public weather and market APIs, and open soil and crop datasets. Steps: collect data, train the crop recommendation model, build the app, then pilot test with local farmers.
- **Scalability:** Starts with one district and expands to new regions and crops by adding local data. The cloud backend handles more users, and features like regional languages, voice support, and SMS alerts can reach farmers with low internet access
- **Risks:** Inaccurate forecasts or missing data, limited internet access, and low farmer trust. Mitigations: use multiple data sources, add offline and SMS support, keep the app simple and multilingual, and validate recommendations with local agriculture experts.

Cost for Product (Optional)

- About 15,000 (one-time). Software is open source and student-built, so this covers the Play Store fee, paid API access, and pilot testing. No hardware is needed.
- Hosting or maintenance cost:About 2,000 per month (24,000 per year) for cloud hosting, database, and API usage, plus minor updates.
- Estimated total:About 40,000 Aprox

Use Cases (compulsorily 3 use cases must be there)

- case 1: Small farmer choosing a crop. A farmer with a 2-acre rain-fed plot enters their location and soil type before the season. The app shows the best crop based on the weather forecast and soil, and tells them when to plant
- case 2:Farmer avoiding oversupply. A farmer planning to grow tomatoes sees from the geographical report that most nearby farms are planting the same crop. The app suggests an alternative that is in higher demand and less crowded locally, so they get a better price.
- case 3: Farmer deciding when and where to sell. After harvest, a farmer checks live market prices and demand. The app recommends the best time and nearest market to sell, reducing transport costs and losses.
  Note: Additional slides may be included only for this section.

Impacts and Benefits

- Farmers get clear, location-specific guidance on what to plant, when to plant, and where to sell. This means fewer crop failures, less guesswork, and better profit.
- Socially, it supports small farmers and strengthens food security. Economically, it raises farm income and reduces local oversupply and price crashes. Environmentally, it cuts wasted seed, water, and fertilizer and promotes crop diversity suited to the climate.
- Target a 15-20 percentage increase in farmer income, a 20-30 percentage drop in crop failure and unsold produce, and 500+ farmers reached in the pilot district within the first year

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG 2	Target 2.3	Zero Hunger	Helps small farmers raise productivity and income by recommending profitable, in-demand crops.
SDG 13	Target 13.1	Climate Action	Helps farmers adapt to unpredictable, climate-driven weather by guiding crop choices.

Individual Member Contribution

Member	Contribution or responsibility
Shweta R T	Leader, content, ideas and implementation
Kanishka S	content, ideas and implementation
Soma Shanmukha Priya Roy	content, ideas and implementation
Tanisha S	content, ideas and implementation

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